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A typed substrate for reproducible computational materials science

Collaboration proposal · G. Barbalinardo, UC Davis

The same physics, a dozen codes — and they don't agree

Lattice thermal conductivity of one crystal, computed by kaldo, phono3py, ShengBTE, LAMMPS — each with its own inputs, units, gauge, and conventions.

Comparing two of them is expert, manual work. Most published computational results never reproduce.

The physics is buried in the code

Units, broadening, Brillouin-zone sums, array layouts, gauge choices — the meaning lives in scripts and tribal knowledge, not in any shared, checkable form.

So reconciliation and reproducibility stay artisanal.

Make the physics the invariant; the code a representation

A typed graph: nodes are physical quantities (ω , D , Γ , c , κ ...), edges are the symbolic formulas relating them. Each code is an **adapter** that pins its outputs onto the shared layer, units and conventions declared.

$$\kappa^{\alpha\beta} = \frac{1}{V} \sum_{q\nu} c_{q\nu} v_{q\nu}^{\alpha} v_{q\nu}^{\beta} \tau_{q\nu}$$

Reconciliation becomes mechanical

- Gauge-invariant **observables** must agree across codes; gauge-dependent intermediates need not — a typed distinction.
- Units reconcile automatically from declared dimensions.
- Cross-code comparison is a structural operation on the graph, not a feat of translation.

It already runs — not a proposal for vaporware

A working operator + representation framework, with:

7 codes mapped

kaldo · 32 variables

phono3py · 31

ShengBTE · 20

phonopy · 17

GPUMD · 8

LAMMPS · 8

ASE · 1

validation engine (derive · compose · cross-check) · automatic units · 600+ tests · an interactive map of the whole graph.

Real cross-code result

Silicon, Tersoff potential, $8\times 8\times 8$ mesh, 300 K — computed through the framework:

RTA **kaldo** 19.46 · **phono3py** 16.74

direct **kaldo** 26.91 · **phono3py** 24.30 W/m·K

The ~15% gap is real — and the substrate **localizes** it: the broadening convention on the scattering rates (Gaussian vs tetrahedron). Reproducibility you can audit, not argue about.

An open, forkable database of physics

Variables + formulas + attached values (simulations and measurements), reconciled at the observables. `git clone` to run locally; pull request to append.

Storing computational physics the way git stores code. [explore the live map](#) →

And the same substrate is what scientific AI will need

Agents that act on typed physics instead of text; an invalid step becomes a **type error**, not a silent wrong number; redundant routes are a built-in verifier. The horizon — and lattice transport is the proving ground.

Let's build it together

MIT brings

domain depth in phonon transport, methods and codes, benchmark systems, students.

We bring

the typed/symbolic substrate, the design, and a working, tested framework.

Together

harden it into a community standard, extend codes and domains, seed the open database, publish the methodology.

Concrete first steps

- Ingest your group's code(s) as adapters onto the shared layer.
- Reproduce a shared benchmark across codes, on the substrate, and pin where they diverge.
- Co-author the cross-code reconciliation methodology (a paper).
- Co-own and grow the open project.

The standard for computational physics — built together.

explore the live map → · openmaterials.ai · G. Barbalinardo, UC Davis